

EE186 Introduction to Embedded Systems - Fall 2026

Final Project

1 Final Project Requirements

Students must work in a group of 2 to 4 people, depending on the scope of the project. The final project must apply the concepts and skills you have learned over the quarter to a problem or application that is **different** from what you have done so far in labs. Your final project must demonstrate your ability in **at least three** of the following categories:

1. Bus interfacing
2. Serial communication
3. Interrupts
4. Interfacing with peripherals
5. Analog-to-digital and/or digital-to-analog conversion

Each group project is allocated **\$30 to purchase additional components for their project**. We will order components after project proposal submission.

2 Schedule

- **Project Idea Submissions Due Date:** October 13th, 2026
- **Project Selection and Team Formation:** October 15th, 2026
- **Project Advising Period:** Once you have formed a team, schedule a meeting with the teaching staff to discuss your project and ensure that it meets all requirements. Advising will take place during Week 5.
- **Final Proposal Submission:** October 27th, 2025
- **Poster Session and Demo:** December 11th, 2026 (12:15 PM - 3:15 PM)

3 Grading

50% - Successful completion.

- Your project has achieved what has been specified in your final proposal
- Your project represents at least three of the lab topic categories listed above

40% - Difficulty components

10% - Final Documentation and Presentation

4 Choosing a Project

Your final project must apply the concepts and skills you have learned during the course to a problem or application that is *different* from what you have done in lab so far. To earn difficulty components, your project must demonstrate your ability to apply your knowledge in the categories mentioned above to an application that is different from labs.

One simple example is a digital alarm clock. Alarm clock functionality includes keeping and displaying the current time, setting the time, setting the alarm, snooze (alarm delay), and alarm activation. All of these functionalities can be realized with your lab kit hardware. This project would demonstrate your competence in bus interfacing, interrupts, and interfacing with peripherals. However, in its most basic form, it does not go beyond what you have already completed in labs. To meet difficulty requirements, you could consider adding an alternate display device, adding a “clap” activated snooze feature.

5 Proposal

The proposal is a general specification of your project at the time of the proposal due date. It should contain the four parts of the general specification listed below. The proposal gives us a concrete, complete plan to evaluate for adequacy and feasibility.

Your proposal will be reviewed for feasibility and adequacy. Course staff will review the projects with each group during the week the proposals are due. A sign-up sheet will be posted for 15-minute appointments.

5.1 General Specification

The general specification is a high-level but complete description of your project. It should be one to two pages long, and include the following four sections.

1. Identify and describe the functions of your project

Example:

- Alarm Clock Functions
- Set Time
- Set Alarm
- Ten-minute Snooze
- Alarm Activated by Turning on Buzzer

2. Specify I/Os and their functions in your project

Example:

- DIP switches: used to enter time
- Push button 1: used to load time
- Push button 2: used to set snooze
- Toggle switches: used to set modes

3. Component or Functional Block Diagram: a diagram that illustrates all functionalities and interfaces of the proposed system

4. Identify any additional components needed for the project (capped at \$30)

6 Implementation

1. Identify tasks within the project. For example, tasks might include the following:

Hardware:

- Develop circuit and wiring diagram

Software:

- Write interrupt handler
- Write I/O interfaces for switches, sensors, displays, etc.
- Program timers

2. Distribute tasks to group members based on preference and ability.
3. Integrate tasks and test the system.

7 Demonstration and Documentation

You will create a poster and demonstrate your project on December 11th, 2026 with all group members present. All group members should be prepared to answer questions on the project.

Details of required documentation will be released in the upcoming weeks. All codes must be submitted.